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THENI

COMMON QUARTERLY EXAMINATION - 2025

Standard XII

Reg.No.

PHYSICS

1919

Time : 3.00 hrs **Part - I** *Marks : 70*

I. Choose the correct answer: **15 x 1 = 15**

1. Fraunhofer lines are an example of _____ spectrum.
   a) line emission b) line absorption
   c) band emission d) band absorption
2. When the current changes from +2A to –2A in 0.05 s, an emf of 8V is induced in a coil. The co-efficient of self induction of the coil is
   a) 0.2 H b) 0.4 H
   c) 0.8 H d) 0.1 H
3. Which charge configuration produces a uniform electric field?
   a) point charge b) uniformly charged infinite line
   c) uniformly charged infinite plane d) uniformly charged spherical shell
4. When the light travels from rarer to denser medium, the angle of incidence and angle of deviation are 60° and 20° respectively, then the angle of refraction is
   a) 80° b) 60° c) 40° d) 20°
5. The temperature co-efficient of resistance of a wire is 0.00125 per °C. At 20°C, its resistance is 1Ω. The resistance of the Wire will be 2Ω at
   a) 800°C b) 850°C
   c) 820°C d) 700°C
6. The vertical component of Earth's magnetic field at a place is equal to the horizontal component. What is the value of the angle of dip at this place?
   a) 30° b) 45° c) 60° d) 90°
7. Two identical conducting balls having positive charges q_1 and q_2 are seperated by a centre to centre distance r. If they are made to touch each other and then separated to the same distance, the force between them will be
   a) less than before b) same as before
   c) more than before d) zero
8. If four cells of 5V emf and 0.5 Ω internal resistance are connected in parallel, then resultant internal resistance of the combination of cells will be
   a) 2 Ω b) 0.125 Ω
   c) 10 Ω d) 2.5 Ω

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9. The resistance of a carbon resistor having colour rings Brown-Black-Orange is
   a) 100 kΩ b) 10 kΩ c) 1 kΩ d) 1000 kΩ
10. If the amplitude of the magnetic field is 3×10^{-6} T, then amplitude of the electric field for a electromagnetic waves is
   a) 100 Vm^{-1} b) 300 Vm^{-1}
   c) 600 Vm^{-1} d) 900 Vm^{-1}
11. The emf induced in the coil 2 is 1.2 V when the rate of flow of current through the coil 1 is 400 As^{-1}. The value of mutual inductance
   a) 3 kH b) 3 H c) 0.03 H d) 3 mH
12. A non-conducting charged ring carrying a charge of q, mass m and radius r is rotated about its axis with a constant angular speed ω. Find the ratio of its magnetic moment with angular momentum is
   a) $\frac{q}{m}$ b) $\frac{2q}{m}$ c) $\frac{q}{2m}$ d) $\frac{q}{4m}$
13. Stars twinkle due to
   a) refraction b) total internal reflection
   c) reflection d) polarisation
14. Which of the following is false for electromagnetic waves?
   a) transverse b) non-mechanical waves
   c) longitudinal d) produced by accelerating charges
15. In a series RL circuit, the resistance and inductive reactance are the same. Then the phase difference between the voltage and current in the circuit is
   a) $\frac{\pi}{4}$ b) $\frac{\pi}{2}$ c) $\frac{\pi}{6}$ d) zero

Part - II

II. Answer any 6 questions. (Q.No.24 is compulsory) **6 x 2 = 12**

16. Mention the ways of producing induced emf.
17. Define capacitance. Give its unit.
18. Compute the speed of the electromagnetic wave in a medium if the amplitude of electric and magnetic fields are 3×10^{4} Nc^{-1} and 2×10^{-4} T respectively.
19. State Ampere's circuital law.
20. Define temperature co-efficient of resistance.

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21. An object is placed at a certain distance from a convex lens of focal length 20 cm. Find the object distance if the image obtained is magnified 4 times.
22. List out the uses of IR rays. (any two)
23. Why do clouds appear white?
24. Compute the magnitude of the magnetic field of a long, straight wire carrying a current of 1 A at distance of 1 m from it.

Part - III

III. Answer any 6 questions. (Q.No.32 is compulsory) **6 x 3 = 18**

25. Write down the properties of electromagnetic waves.
26. Differentiate electric power and electric energy.
27. Consider a point charge +q placed at the origin and another point charge –2q placed at a distance of 9 m from the charge +q. Determine the point between the two charges at which electric potential is zero.
28. How is a galvanometer converted into an ammeter?
29. Explain the various energy losses in a transformer?
30. Find the heat energy produced in a resistance of 10 Ω when 5 A current flows through it for 5 minutes.
31. Define critical angle and what are the conditions for total internal reflection to takes place.
32. Find the impedence of a series RLC circuit If the inductive reactance, capacitive reactance and resistance are 184 Ω, 144 Ω and 30 Ω respectively. Also calculate the phase angle between voltage and current.
33. Write down the applications of capacitor.

Part - IV

IV. Answer all the questions. **5 x 5 = 25**

34. a) Derive an expression for the electric field due to a dipole on its equatorial plane.

(OR)

b) Obtain len's maker's formula and len's equation.

35. a) Calculate the force on a current carrying conductor in a magnetic field.

(OR)

b) Describe the microscopic model of current and obtain microscopic form of Ohm's law.

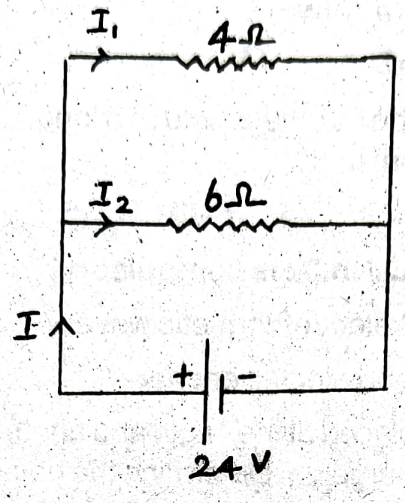
36. a) Show mathematically that the rotation of a coil in a magnetic field over one rotation induces an alternating emf of one cycle.

(OR)

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b) i) Obtain the effective resistance of resistors connected in series.

ii) Calculate the equivalent resistance in the following circuit.



37. a) Explain the types of emission spectrum.

(OR)

b) Obtain a relation for the magnetic field at a point along the axis of a circular coil carrying current using Biot-Savart law.

38. a) Obtain the phase relation between voltage and current in a capacitive circuit.

(OR)

b) i) Obtain the expression for energy stored in a capacitor.

ii) A parallel plate capacitor has square plates of side 5 cm and seperated by a distance of 1 mm. Calculate the capacitance of this capacitor

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